RATIONALS, SEMICIRCLES AND METALLIC RATIOS ON ONE PAGE

From rational radii to metallic ratios

The radius $3/2$ is rational. The height $\sqrt{2}$ shows where a new number enters.

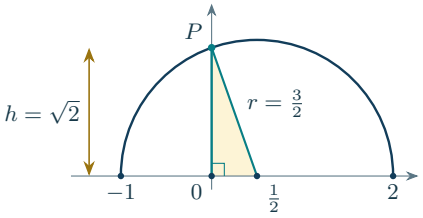
1 / READ THE INNERMOST SEMICIRCLE

Why is the first radius $3/2$?

The diameter runs from $A = (-1, 0)$ to $B = (2, 0)$. Its length is 3, so the centre is $C = (1/2, 0)$ and the radius is $r = 3/2$. Take $O = (0, 0)$ and the upper point $P = (0, h)$ on the circle. The right triangle OCP gives

$$h^2 + \left(\frac{1}{2}\right)^2 = \left(\frac{3}{2}\right)^2, \qquad \boxed{h^2 = 2, \quad h = \sqrt{2}.}$$

The starting coordinates, centre and radius are all rational. The intersection introduces the new height.



2 / WHAT DOES RATIONAL MEAN?

Fractions of integers form $\mathbb{Q}$

Write $\mathbb{Z}$ for the integers. The **rational numbers** are

$$\mathbb{Q} = \left\{ \frac{a}{b} : a, b \in \mathbb{Z},\ b \neq 0 \right\}.$$

For example, $3/2 = 1.5$, $1/3 = 0.\overline{3}$ and $2 = 2/1$ are rational. A repeating decimal can be rational too.

Adding, subtracting, multiplying and dividing rational numbers keeps us in $\mathbb{Q}$, provided we divide by a nonzero number. Taking square roots can enlarge the system: $\sqrt{9/4} = 3/2$ is rational, while $\sqrt{2}$ is irrational.

If $\sqrt{2} = a/b$ in lowest terms, $a^2 = 2b^2$ makes a even. Writing $a = 2k$ gives $b^2 = 2k^2$, so b is even too. This contradicts lowest terms.

3 / MOVE THE RIGHT ENDPOINT TO q

A whole family of rational circles

Take a positive rational q and use diameter endpoints -1 and q. The centre coordinate and radius are

$$c = \frac{q-1}{2}, \qquad r = \frac{q+1}{2}.$$

Both are rational. The circle is $(x-c)^2 + y^2 = r^2$. At $x = 0$, its upper height h satisfies

$$\boxed{h^2 = r^2 - c^2 = \frac{(q+1)^2 - (q-1)^2}{4} = q.}$$

So every upper intercept is $h = \sqrt{q}$.

The radius r is the **arithmetic mean** of $1, q$; the height $\sqrt{1 \cdot q}$ is their **geometric mean**. Since $r^2 - h^2 = c^2 \geq 0$, this also proves $r \geq h$.

4 / ALLOW THE NEW HEIGHT IN THE ARITHMETIC

One square root gives two coordinates

Adjoining $\sqrt{2}$ means allowing rational combinations of the building blocks $1, \sqrt{2}$:

$$K = \mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}.$$

Sums and products stay in this form. For division, multiply top and bottom by $a - b\sqrt{2}$:

$$\frac{1}{a + b\sqrt{2}} = \frac{a - b\sqrt{2}}{a^2 - 2b^2}, \qquad a + b\sqrt{2} \neq 0.$$

Thus K is a **field**: division by any nonzero element stays inside it. Its two independent rational coordinates a, b give this extension of $\mathbb{Q}$ **degree** 2.

Already $1 + \sqrt{2}$ and $\sqrt{8} = 2\sqrt{2}$ belong to K. A different-looking root can use the same number system. At $q = 4$, the height is 2, which already belongs to $\mathbb{Q}$; that circle needs no field extension.

5 / CHOOSE THE RADICAND FOR A METAL

The metallic ratio uses the same new root

For integer $\lambda \geq 1$, choose $q = \lambda^2 + 4$. The semicircle then gives

$$r = \frac{\lambda^2 + 5}{2}, \qquad h = \sqrt{\lambda^2 + 4}.$$

The radius is rational. Halve the height and add the rational length $\lambda/2$:

$$\boxed{\mu_\lambda = \frac{\lambda + h}{2}}, \qquad \mu_\lambda^2 - \lambda\mu_\lambda - 1 = 0.$$

This is the λ-**metallic ratio**. Conversely,

$$h = 2\mu_\lambda - \lambda.$$

Each number can be recovered from the other using rational arithmetic, so

$$\boxed{\mathbb{Q}(\mu_\lambda) = \mathbb{Q}\left(\sqrt{\lambda^2 + 4}\right).}$$

6 / SAMPLE CONSTRUCTIONS

Which rational radius produces each metallic ratio?

λ	metal	$q = \lambda^2 + 4$	radius r	μ_λ	field
1	Golden	5	3	$(1 + \sqrt{5})/2$	$\mathbb{Q}(\sqrt{5})$
2	Silver	8	$9/2$	$1 + \sqrt{2}$	$\mathbb{Q}(\sqrt{2})$
3	Bronze	13	7	$(3 + \sqrt{13})/2$	$\mathbb{Q}(\sqrt{13})$
4	4-metallic	20	$21/2$	$2 + \sqrt{5}$	$\mathbb{Q}(\sqrt{5})$

Silver is already $1 + \sqrt{2}$ from the first circle with $r = 3/2$. The general recipe uses $q = 8$, whose height $2\sqrt{2}$ is halved. Both routes stay in $\mathbb{Q}(\sqrt{2})$. Golden and 4-metallic share $\mathbb{Q}(\sqrt{5})$.

Further reading. Euclid, *Elements*, VI.13, the geometric-mean construction; J. B. Gil and A. Worley, *Generalized Metallic Means*, §1.

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